

# Evaluation Rubric — 100 Points

| Criterion                                       | Points |
|-------------------------------------------------|--------|
| Retrieval Quality                               | 22     |
| Answer Grounding & Faithfulness                 | 18     |
| System Architecture & Full-Stack Implementation | 15     |
| Evaluation & Metrics Implementation             | 10     |
| Clinical Safety & Responsible AI                | 8      |
| Presentation, Communication & Live Demo         | 17     |
| Innovation & Out-of-the-Box Thinking            | 10     |
| Total                                           | 100    |

## 1. Retrieval Quality — 22 points

This criterion specifically evaluates:

- Relevance of the top-k retrieved chunks to the actual clinical question asked not just topically related, but specifically responsive.
- Retrieval Precision@k, measured against a real test set (not eyeballed on one or two example queries).
- The chunking strategy itself: is it section-aware and deliberately chosen, or an arbitrary fixed-size split with no justification?
- Embedding model selection: was more than one model or configuration actually compared, with a documented reason for the final choice?
- Transparency of the retrieval step — are the retrieved chunks shown before generation, so a judge (or a clinician) can verify what evidence the answer is built on?

## 2. Answer Grounding & Faithfulness — 18 points

This criterion specifically evaluates:

- Accuracy of citations: does each citation point to the specific document, section, and page that actually contains the claim not just the same document in general?
- Direct support: is every clinical recommendation in the answer explicitly backed by the cited excerpt, with no unsupported addition?
- Absence of hallucinated claims statements that sound plausible but do not appear anywhere in the retrieved evidence.
- Whether the system preserves the original strength of a recommendation (e.g. a WHO 'conditional' vs 'strong' recommendation) rather than flattening all guidance into the same confident tone.

## 3. System Architecture & Full-Stack Implementation — 15 points

This criterion specifically evaluates:

- Clear separation between the ingestion, retrieval, generation, and safety layers, with a modular, swappable design (e.g. could the vector database be replaced without rewriting unrelated code?).
- Code quality and documentation: is the codebase something another engineer could pick up and maintain?
- Error handling: what happens when PDF parsing fails, a document has complex tables, or retrieval returns nothing?

- Full-stack completeness: is there an actual working user interface connected live to the backend not a static mockup, and not a terminal or notebook standing in for a real demo?

#### 4. Evaluation & Metrics Implementation — 10 points

This criterion specifically evaluates:

- Whether the metrics used are actually appropriate for a RAG system (Precision@k, citation accuracy, faithfulness) rather than generic or irrelevant measures.
- Existence and quality of a real test set: how many questions, and how were the expected/reference answers verified?
- Whether low scores led to a real next step — teams should be able to explain what they changed after seeing a weak metric, not just report the number.
- Awareness of trade-offs, such as retrieval accuracy versus response latency.

#### 5. Clinical Safety & Responsible AI — 8 points

This criterion specifically evaluates:

- Does the system clearly and explicitly refuse to answer when the retrieved evidence is insufficient, rather than guessing?
- Are there visible warnings against over-confidence, and does the interface avoid implying the system is a substitute for a clinician's judgment?
- Is there a defined, tested confidence threshold that gates generation on retrieval quality?
- Can the system correctly distinguish between a question it can answer and a near-identical one that falls outside its indexed sources?

#### 6. Presentation, Communication & Live Demo — 17 points

This criterion specifically evaluates:

- Clarity of the interface: can a busy clinician find the source behind a recommendation in a couple of seconds, without hunting for it?
- Visual distinction between a confident, well-supported answer and one carrying an insufficient-evidence warning.
- Smoothness of the live demonstration itself does it run without technical failures, and does the team have a rehearsed backup plan if it does?
- Narrative clarity: can the team explain the problem, the solution, and the result in plain language a non-technical judge can follow, not just a technically accurate but jargon-heavy description?

#### 7. Innovation & Out-of-the-Box Thinking — 10 points

This criterion specifically evaluates:

- Did the team approach the clinical problem in a genuinely non-obvious way compared to a standard RAG implementation?
- Do any additional techniques used (re-ranking, multi-agent reasoning, hybrid search, etc.) solve a problem that existed in the baseline system, with evidence that it helped?
- Is the innovation explainable and defensible under questioning, or does it collapse under a follow-up question about why it was necessary?
- This is now a fully scored category, not optional bonus credit — a team that skips it is leaving 10 real points on the table.